

SHUBHANGI THAKUR

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LinkedIn GitHub

EDUCATION

JAGRAN LAKECITY UNIVERSITY

Bachelor of Technology in Computer Science and Engineering

Bhopal, Madhya Pradesh

June 2023 – Present

Specialization in Artificial Intelligence and Machine Learning

CGPA: 9.44/10.00

EXPERIENCE

Open Source Contributor – GSSoC'26

May 2026 – Present

Remote

- Contributing to open-source software projects through GitHub-based development, issue resolution, documentation, and collaborative workflows.
- Collaborating with maintainers and contributors to understand requirements, implement changes, troubleshoot issues, and improve project functionality.
- Applying software engineering practices including version control, debugging, code organization, and code review.

TECHNICAL SKILLS

Programming: Python, Java, C/C++, JavaScript, SQL

Generative AI: LLMs, Prompt Engineering, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, AI Agents, Embeddings, Semantic Search

Machine Learning: Machine Learning, Deep Learning, NLP, Computer Vision, Explainable AI, Reinforcement Learning, Predictive Modeling

Frameworks & Libraries: FastAPI, Django, Flask, React.js, PyTorch, TensorFlow, Scikit-learn, OpenCV, XGBoost

Databases & AI Infrastructure: PostgreSQL, Qdrant, FAISS, Neo4j, Redis, SQLite, Firebase, Docker

Development Tools: REST APIs, Git, GitHub, Celery, Streamlit, VS Code, Jupyter Notebook

PROJECTS

Explainable Bilingual Multi-Agent Legal Intelligence Platform

Ongoing | July 2026 – Present

Multi-Agent AI, LLMs, RAG, Knowledge Graphs, Qdrant, Neo4j, FastAPI, OCR

- Developing a bilingual legal intelligence platform for processing Hindi, English, and mixed-language legal documents including judgments, petitions, FIRs, and affidavits.
- Designing a multi-agent AI workflow for document classification, legal provision identification, case-law retrieval, and context-aware response generation.
- Implementing Retrieval-Augmented Generation (RAG) using Qdrant for semantic document retrieval and Neo4j for relationship-based legal knowledge representation.
- Integrating OCR-based document processing to extract information from scanned legal documents before downstream classification and retrieval.
- Building explainable AI workflows that associate generated responses with contextual evidence and retrieved legal information to improve transparency.
- Developing FastAPI-based REST services for document processing, OCR, agent orchestration, retrieval, and AI response generation.

Autonomous Surgical Decision System

January 2026 – June 2026

Machine Learning, Computer Vision, NLP, Predictive Modeling, Explainable AI

- Built an AI-driven clinical decision support system for surgical risk prediction and continuous patient monitoring across multiple stages.
- Designed a multimodal machine learning pipeline combining NLP, computer vision, and predictive modeling for processing heterogeneous clinical information.
- Developed machine learning workflows for risk prediction and decision support using structured and unstructured input data.
- Applied SHAP, LIME, and Grad-CAM to interpret model predictions and provide transparent insights into model decision-making.

Sugarcane Disease Detection System

June 2025 – December 2025

Computer Vision, Deep Learning, CNNs, TensorFlow/PyTorch, OpenCV

- Built a deep learning image classification system to automate sugarcane disease detection using Convolutional Neural Network (CNN) architectures.
- Developed an end-to-end computer vision pipeline covering image preprocessing, dataset preparation, model training, evaluation, and prediction.
- Implemented image processing and augmentation workflows using OpenCV to improve the quality and consistency of model inputs.
- Used TensorFlow/PyTorch for deep learning model development and generated disease predictions to support agricultural monitoring and decision-making.

ACHIEVEMENTS

- Maintained a **9+ CGPA** and received a **Merit Certificate** for academic excellence.
- Participated in **8+ hackathons**, developing AI/ML-based solutions for real-world problems.
- Presented Machine Learning research at **RAEMPS'2025** and **RICCE'2024** conferences.